 # Operational Memo -- Scenario C: Governance Across 14, 6, and 0 Crew Regimes

****To:** SDV Program Office & Vessel Operations Command**
From: Principal Systems Architect, ‘sdv-resource-governor‘
Subject: Deterministic Resource, Deck-Safety, and Autonomy Rules Under Crew Regimes **14**, **6**, and **0**

1. Executive Summary
A Software-Defined Vessel (SDV) cannot treat human crew availability as a mere scheduling annotation: crew count directly governs **which physical payload transitions are admissible**, **how many deck evolutions can execute concurrently**, **whether at-sea vehicle replacement ($A’$) is physically safe**, and **how fault recovery escalates when telemetry degrades or a payload overconsumes**. In ‘sdv-resource-governor‘, human crew availability is enforced as a hard operational guard inside ‘scheduling.py‘ and ‘admission.py‘.

2. Quantitative Regime Comparison (‘FULL_CREW_14‘ vs. ‘REDUCED_CREW_6‘ vs. ‘UNCREWED_0‘)

Operational Dimension	Regime 1: **14 Crew** (‘FULL_CREW_14‘)	Regime 2: **6 Crew** (‘REDUCED_CREW_6‘)	Regime 3: **0 Crew** (‘UNCREWED_0‘)
Bridge & Engine Watch	‘5‘ personnel	‘4‘ personnel	‘0‘ (Remote Shore Control + Onboard Autonomy)
Available Deck/Payload Crew	**‘9‘ personnel**	**‘2‘ personnel**	**‘0‘ personnel**
Max Concurrent Manual Deck Ops	**‘2‘ simultaneous** (e.g., G2 ROV + G4 UAV)	**‘1‘ serialized** (G2 *or* G4, never simultaneous)	**‘0‘ manual deck evolutions**
G4 UAV Launch / Recovery (‘5-12 kn‘, wind $\le 25\text{ kn}‘, ‘SS <= 4‘)	**PERMITTED** (‘15-60 s‘ local bridge confirmation)	**PERMITTED** (‘1-3 min‘ watch handover confirmation; blocks G2 deck ops)	**FORBIDDEN** if manual deck arrest/handling required; only pre-stowed or automated modes
G2 Winch / Probe Deployment (‘0-5 kn‘, wind $\le 20\text{ kn}‘, ‘SS <= 4‘)	**PERMITTED**	**PERMITTED** (serialized; cannot overlap G4 flight deck ops)	**FORBIDDEN** (requires umbilical/winch deck handling)
At-Sea Vehicle Replacement ($A’$) (‘G4‘: ‘3‘ crew / ‘30 min‘; ‘G3‘: ‘4‘ crew / ‘45 min‘)	**PERMITTED** (‘available_crew=9 >= 4‘ required for crane/hangar replacement at sea)	**BLOCKED** (‘available_crew=2 < 3‘ for G4 and ‘< 4‘ for G3 safe replacement at sea)	**FORBIDDEN** (‘0‘ crew; all physical replacements deferred to port)
At-Sea Physical Module Install (‘G6‘ ‘10 ft‘ lab, ‘min_crew=4‘)	**PERMITTED** in calm $A’$ window (‘0-4 kn‘, wind $\le 12\text{ kn}$)	**BLOCKED** (‘available_crew=2 < 4‘)	**FORBIDDEN** (‘CRW-02‘: zero physical installation changes at sea)

| **Autonomous Sensor & Compute Guests** (‘G1‘, ‘G3‘ in-water survey, ‘G5‘, ‘G6‘ auto-sampler) | **100% Operational** | **100% Operational** | **100% Operational** (governed via software mode switching, throttle, and power shed) |

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3. Acoustic Survey Deck-Machinery Lockout & $A’$ Window Separation

1. **Acoustic Survey Lockout (‘G3‘ ‘forbids_deck_machinery_while_active=True‘)**:
   - Regardless of whether **14** or **6** crew are aboard, while **G3** is in ‘SURVEY_QUIET‘ (‘1 kW‘, ‘2-6 kn‘), hydraulic A-frames, winches (‘G2‘), and flight-deck handling gear (‘G4‘) are **locked out** to prevent acoustic corruption of the seabed survey sonar.
   - To launch or recover **G4** (‘UAV_SAR‘) or deploy **G2** (‘ROV_INSPECTION‘), the governor must first issue ‘ACOUSTIC_HOLD‘ to **G3**, wait for the mandatory **‘15 s‘ acoustic modem handshake** (‘10-60 s‘ spec), and confirm G3 is in ‘HOLD_ACOUSTIC‘ before unlocking deck machinery or turning into the wind.
2. **Separation of Normal Flight Window vs. $A’$ At-Sea Replacement Window (‘CRW-03‘)**:
   - **Normal G4 Flight Launch/Recovery**: vessel speed **‘5-12 kn‘**, wind **$\le 25\text{ kn}$**, Sea State **≤ 4**, heading aligned within $\pm 15^\circ$ of wind, **‘2‘ deck crew**.
   - **$A’$ At-Sea Vehicle Replacement**: vessel speed **‘4-8 kn‘**, wind **$\le 20\text{ kn}$**, Sea State **≤ 3** (‘G4‘, ‘3‘ crew / ‘30 min‘) or **≤ 4** (‘G3‘, ‘4‘ crew / ‘45 min‘). At ‘10 kn‘ and ‘22 kn‘ wind, normal UAV recovery is admitted (‘YES‘), whereas at-sea vehicle replacement is rejected (‘NOT_NOW‘ / blocked by $A’$ envelope).

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4. Fault Governance Under Reduced and Uncrewed Regimes (‘RUN-03‘, ‘RUN-04‘, ‘SAF-01‘)

When operating with **6** or **0** crew, physical manual intervention on tripped breakers or overheated racks is slow (**6 crew**) or impossible (**0 crew**). The governor enforces three automated fail-safe barriers:

- **Urgent Safety Commands (‘STOP‘, ‘HOLD‘, ‘SECURE‘ $\le 200\text{ ms}$, ‘SAF-01‘)**: Routed strictly over the dual ‘10 Gbps‘ local shipboard LAN (‘<5 ms‘ transport latency), never over ‘600-800 ms‘ SATCOM. Measured execution latency on both MacBook Pro and the ‘aarch64‘ Linux board (‘training06‘) is **‘< 1 ms‘** ($\ll 200\text{ ms}$ budget).
- **Guest Overconsumption (‘+10%‘ to ‘+20%‘, ‘RUN-04‘)**: Because none of ‘G1‘-‘G5‘ is safety-critical by design, if a payload (e.g., ‘G5‘ declared ‘45 kW‘ drawing ‘54 kW‘ = ‘+20%‘) exceeds its manifest envelope, 1:1 thermal coupling immediately adds ‘+9 kW‘ to the chilled-water ledger and the governor autonomously throttles ‘G5‘ to ‘DEGRADED_70PCT_LOSS‘ (‘8 kW‘) while protecting active uninterruptible/recovery tasks (‘G3‘ recharge, ‘G4‘ flight/recovery).
- **Lost or Stale Telemetry Feeds (‘RUN-03‘)**: If ‘seawater_temp_c‘ exceeds its ‘5,000 ms‘ TTL (‘STALE‘ / ‘UNAVAILABLE‘), the governor never extrapolates optimistic cooling; it clamps to the conservative **$30\,^\circ\text{C}$ (‘120 kW‘ nominal / ‘72 kW‘ degraded)** thermal bound until sensor freshness is restored.